MCO FINX: Financial Information Retrieval for Opinion-Aware Document Ranking

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Abstract

Our project addresses opinion-based information retrieval in the financial domain, focusing on ranking relevant documents for natural language queries that express evaluation, uncertainty, and sentiment. We study the FiQA benchmark collection, which consists of financial microblogs, community question–answer posts and short news-like texts, paired with binary relevance judgments. To handle the narrative and sentiment-rich nature of the data, we propose a hybrid retrieval approach that combines lexical retrieval with BM25, dense semantic similarity using Sentence-BERT embeddings and selective sentiment analysis based on FinBERT. Sentiment signals are applied only to opinion-oriented queries to avoid introducing noise. We evaluate our systems using standard IR metrics including MAP, Precision at k, Recall at k, and nDCG. Experimental results show that dense semantic retrieval significantly outperforms lexical baselines, and that selectively integrating sentiment information further improves ranking quality for opinion-driven financial queries.

1 Introduction and Motivation

This project addresses the problem of retrieving and ranking relevant documents for domain specific financial queries that are often opinion driven, evaluative, and sentiment laden. In modern financial information environments, users frequently seek explanations, reasoning, and perspectives rather than isolated factual answers. Such information need requires retrieval systems to go beyond keyword matching and capture semantic meaning and sentiment polarity.

The intended users include retail investors, financial analysts, fintech developers, researchers, and participants of online financial discussion communities. For these users, timely access to relevant and trustworthy information is essential, as investment decisions are strongly influenced by market sentiment, perceived risk, and collective opinion. Improving retrieval quality in this setting supports better interpretation of financial discourse and more informed decision making.

The domain context of this work is finance, where language is specialised, informal, and rich in subjective evaluation. The research question guiding this project is: *How can a hybrid retrieval system combining lexical relevance, semantic similarity, and sentiment polarity improve the ranking of opinion oriented financial queries compared to purely lexical or purely semantic approaches in the FiQA dataset?*

To address this research question, we propose a hybrid retrieval pipeline in which candidate documents are first retrieved using BM25 for high lexical recall, then refined through a FinBERT-based bi-encoder to capture semantic relevance, and finally enhanced with a selective sentiment classification stage that is applied only to opinion-oriented queries in order to align document polarity with the user's evaluative intent.

2 Task and Dataset Description

The task that we studied in this project is opinion based financial information retrieval. Given a natural language query, the system must retrieve and rank documents that provide relevant explanations, viewpoints, or opinions, rather than factual answers. Relevance therefore depends on semantic alignment and underlying opinion or sentiment expressed in the documents.

We use the FiQA dataset, a benchmark collection for opinion based financial question answering, consisting of approximately 57 thousands of documents and 648 queries. The documents originate from financial microblogs, community question answer platforms, and short news like texts, reflecting real world online financial discourse. Our exploratory analysis showed that 17 percent of the queries contain explicit sentiment or evaluative language, such as references to risk, value, or comparative judgement. This observation confirms that a substantial portion of financial information needs in FiQA cannot be addressed through lexical matching alone.

Additionally, the dataset is marked by domain specific vocabulary and limited numeric content, indicating that the primary challenge lies in interpreting narrative and opinion oriented financial language rather than numerical reasoning. These characteristics make FiQA a challenging benchmark for traditional IR approaches and motivate the use of hybrid retrieval methods that integrate semantic understanding and sentiment aware signals, in line with findings reported in *Information Retrieval in Finance: Industry and Academic Perspectives on Innovation*.

3 Methodology

3.1 Baseline Systems (Phase I)

To establish reference points for evaluating more advanced retrieval strategies, we implemented three baseline systems of increasing complexity. These baselines allow us to isolate the contribution of lexical matching, query expansion, and lightweight entity awareness before introducing semantic and sentiment-based components.

Baseline 1: BM25 (Lexical Retrieval)

The first baseline uses BM25 as a pure sparse retrieval model based on keyword overlap between queries and documents. This run establishes the fundamental performance level achievable through lexical matching alone and serves as a reference point for all subsequent experiments.

The BM25 baseline achieves reasonable early precision and ranking quality, indicating that it effectively matches core financial terms and entities present in the queries. Recall improves at higher cutoffs, while precision decreases, reflecting the sparsity of relevance in the dataset. However, the absence of semantic understanding and sentiment modeling limits performance on opinion-driven queries.

Baseline 2: BM25 + RM3 Query Expansion

The second baseline augments BM25 with RM3 to test whether query expansion improves retrieval for short and ambiguous financial queries. The motivation is that expanded queries might capture additional relevant terms not explicitly mentioned in the original query.

Query expansion with RM3 does not improve overall effectiveness compared to the BM25 baseline. Both MAP and early precision slightly decrease, while recall remains largely unchanged. This suggests that pseudo-relevance feedback introduces noisy or weakly related expansion terms, which is particularly harmful in short, opinion-oriented financial queries. These results indicate that lexical expansion alone is insufficient and motivate the use of semantic or sentiment-aware signals.

Baseline 3: BM25 + Simple Entity Matching

The third baseline introduces a lightweight entity-aware reranking strategy. In this approach, the BM25 score is augmented with an entity overlap signal capturing shared company names or stock tickers between the query and the document. A weighting parameter λ controls the influence of the entity signal relative to the BM25 score.

Entity-aware reranking yields small but consistent improvements when the entity signal is moderately weighted. Very small values of λ slightly reduce effectiveness, indicating that a weak entity signal adds little benefit. A moderate value of $\lambda = 0.5$ provides the best balance, improving recall and nDCG without harming precision. Larger values do not lead to further gains and may partially offset lexical relevance. These results confirm that entity matching is useful as a complementary signal, but not as a dominant ranking factor.

3.2 Advanced Systems (Phase II)

In Phase II, we move beyond purely lexical and lightweight enhancements and explore neural and hybrid retrieval strategies designed to better capture semantic meaning and sentiment in opinion oriented financial queries. These advanced systems explicitly test whether dense semantic representations and selective sentiment modeling can address the limitations observed in the baseline runs.

E1 — Dense Semantic Retrieval (Bi-Encoder)

Experiment E1 introduces a dense semantic retriever based on a bi-encoder architecture using Sentence BERT. Unlike BM25, this model ranks documents by semantic similarity between query and document embeddings. The hypothesis tested in this experiment is that dense representations better capture the meaning of financial questions and explanatory answers, particularly when wording differs between queries and relevant documents.

The system is split into offline and online components. Offline, document embeddings are computed once using the bi-encoder and stored in a dense index built with FlexIndex. This step is computationally expensive but query independent. Online, each query is encoded at runtime using the same model, and nearest neighbor search is performed in the dense index to retrieve the most semantically similar documents.

The dense semantic retriever significantly outperforms the BM25 baseline across all evaluation metrics. Mean Average Precision increases from 0.21 to 0.31, and both precision and recall at low cutoffs improve substantially. These gains indicate that dense semantic representations are more effective at matching opinion rich financial queries with relevant explanatory documents. The strong performance of E1 confirms the hypothesis that semantic similarity is crucial for this task and motivates its use as a core component in the final hybrid system.

E2 — Hybrid Retrieval: BM25 + Dense Semantics + Selective Sentiment Boost (Final System)

Experiment E2 extends E1 by integrating lexical relevance, dense semantic similarity, and sentiment polarity in a selective hybrid retrieval pipeline. The key hypothesis is that sentiment can improve ranking for opinion-oriented queries but should not be applied uniformly, as this may introduce noise. Therefore, sentiment is activated only when a query contains explicit evaluative cues.

The E2 pipeline consists of four stages. First, BM25 retrieves a top-K set of candidate documents to ensure high recall. Second, candidates are reranked using dense semantic similarity computed with a Sentence-BERT bi-encoder. Third, a lightweight query analysis determines whether the query is opinion-oriented; only in this case is a FinBERT sentiment classifier applied. Finally, lexical, semantic, and optionally sentiment scores are combined into a single ranking score, with sentiment assigned a small weight as a refinement factor.

Query-independent resources, including the dense document index and document-level sentiment scores, are precomputed offline to ensure efficiency. E2 substantially improves over BM25 across all metrics, increasing precision and ranking quality while preserving recall,l confirming that selectively weighted sentiment information provides complementary benefits when combined with lexical and semantic retrieval.

Implementation Details

All experiments are implemented using PyTerrier, which provides support for indexing, retrieval pipelines, and evaluation, with Terrier as the underlying sparse retrieval engine. Neural components are implemented using Hugging Face Transformers, including Sentence-BERT, FinBERT for dense semantic retrieval and sentiment classification. Data handling and analysis are performed using Pandas.

Before retrieval, we apply a minimal query sanitisation step to ensure compatibility with the Terrier query parser. This step removes characters that may cause parsing issues, such as quotation marks and colons, and normalises whitespace. The preprocessing is purely technical and does not modify the semantic content of the queries.

The FiQA corpus is indexed from scratch using PyTerrier's IterDictIndexer, indexing document text and storing document identifiers as metadata. Basic integrity checks confirm that missing or empty documents represent a negligible fraction of the corpus and do not impact retrieval results.

4 Experiments and Results

4.1 Evaluation Setup

We evaluate all retrieval systems using standard Information Retrieval metrics, including Precision at k (P@1, P@5, P@10), Recall at k (R@5, R@10), nDCG@5 and nDCG@10, and Mean Average Precision (MAP). Relevance judgments are provided by the FiQA qrels and are binary, with documents labeled as relevant or non-relevant. All systems are evaluated against the same qrels to ensure fair and reproducible comparison. FiQA represents a sparse relevance setting, with most queries having only one or two relevant documents. This increases the difficulty of the task, as systems must accurately identify a very small set of relevant documents among many non-relevant ones.

4.2 Results

Relative Improvements and Their Causes

The experimental results show clear and consistent improvements as the retrieval pipeline moves from purely lexical methods to semantic and hybrid approaches. The BM25 baseline establishes a solid reference point, demonstrating that lexical matching is effective at identifying core financial terms and entities. However, its performance is limited by an inability to capture semantic similarity and subjective intent.

Introducing dense semantic retrieval in E1 (Bi-encoder Sentence-BERT) leads to a substantial improvement across all metrics. MAP increases markedly, and both precision and recall at low cutoffs improve, indicating that dense embeddings better capture the meaning of opinion-oriented financial queries and explanatory answers. This confirms that semantic similarity is a crucial factor for this task, especially when relevant documents use different wording than the query.

The final hybrid system E2, which combines BM25, dense semantic similarity, and selective sentiment boosting, achieves the best overall balance between ranking quality and robustness, confirming that sentiment information can act as a useful refinement signal when applied selectively. Precision-oriented metrics and nDCG benefit from integrating lexical and semantic signals, while recall remains high due to the BM25 first-stage retrieval. achieves the best overall balance between ranking quality and robustness, confirming that sentiment information can act as a useful refinement signal when applied selectively.

	name	map	P_1	P_5	P_10	recall_5	recall_10	ndcg_cut_5	ndcg_cut_10
0	BM25	0.210385	0.236111	0.106481	0.070370	0.247471	0.309708	0.230060	0.252589
1	BM25+RM3	0.206532	0.220679	0.107099	0.067593	0.243568	0.303302	0.225140	0.245375
2	BM25 + EntityBoost (TOPK=100, λ=0.5)	0.210495	0.237654	0.107099	0.069753	0.250104	0.310195	0.232265	0.253808
3	E2 Hybrid: BM25@100 + Dense (SBERT) + FinBERT ...	0.322441	0.370370	0.169444	0.104167	0.377535	0.448816	0.361309	0.383887

Failures, Biases, and Inconsistencies

While dense retrieval performs strongly, it can overemphasize semantic similarity and occasionally retrieve documents that are topically related but not strictly relevant. In the hybrid system, sentiment boosting introduces mild noise when averaged across all queries, as sentiment is useful only for a subset of opinion-oriented questions, explaining why the untuned hybrid model does not initially outperform pure dense retrieval in MAP. Query expansion with RM3 consistently underperforms due to noisy pseudo-relevance feedback, and simple entity matching yields only marginal gains, confirming that lexical enhancements alone are insufficient without deeper semantic modeling.

What Worked Well for the Financial Domain

The results highlight that semantic representations are particularly effective for financial QA tasks involving explanations, reasoning, and opinion. Dense embeddings successfully bridge lexical gaps and capture nuances of financial language. Moreover, selective sentiment modeling, when carefully weighted and applied only to opinion-oriented queries, improves ranking quality without harming factual retrieval.

Overall, the experiments demonstrate that financial information retrieval benefits most from hybrid systems that combine lexical precision, semantic understanding, and sentiment awareness. This aligns well with the narrative and opinion-rich nature of the FiQA dataset and confirms the suitability of the proposed approach for opinion-based financial retrieval.

5 Discussion and Conclusions

This project helped us understand that **semantic retrieval is the key driver of performance** for opinion-oriented financial queries. Dense SBERT representations consistently outperformed lexical baselines, showing that relevant documents often match the underlying intent rather than exact query terms. In contrast, techniques such as RM3 query expansion and simple entity overlap produced only limited improvements.

A major challenge during implementation was integrating sentiment information in a way that was both computationally efficient and methodologically sound. Applying FinBERT uniformly at query time would have been prohibitively expensive and risked introducing noise into the ranking. To address this, we had to carefully design an offline computation and caching strategy for document-level sentiment scores, as well as a gating mechanism to ensure that sentiment was applied only when appropriate.

From this experience, we learned the importance of balancing retrieval signals instead of adding complexity indiscriminately. Future work could focus on replacing rule-based query cues with a more advanced learned query intent classifier, as well as exploring stronger reranking models and more fine-grained entity or aspect-aware representations.

Future work could focus on replacing the rule-based query cues with a learned query intent classifier to more accurately distinguish between factual and opinion-oriented queries. Additionally, while we performed limited sensitivity analyses, a more extensive hyperparameter tuning in Experiment E2 could further improve performance. However, this was intentionally avoided due to its high computational cost relative to the scope and objectives of the project.

Takeaway: dense semantic retrieval provides a strong foundation for financial opinion search, while sentiment can offer additional value when applied selectively and with care.

AI Tools and Assistance Declaration (Required)

This project was developed with the support of AI-based tools used **exclusively as assistive instruments** during the research and writing process. In particular, **ChatGPT 5.2** was used to support brainstorming, clarify methodological choices, and refine technical explanations.

AI assistance was also used to help structure experimental descriptions, interpret intermediate results, and improve English phrasing. All system design decisions, experimental implementations, evaluations, and interpretations were conceived, executed, and validated by the authors.

No AI tool was used to generate experimental results, automatically write code without supervision, or replace critical reasoning, analysis, or original contributions by the team. All responsibility for the content, correctness, and conclusions of this work lies with the authors.